Lab Report

Introduction

The objective of this experiment is to investigate the generation of the skin effect on conductor surfaces and its relationship with frequency and material parameters through theoretical calculations and COMSOL simulations.

The experimental procedure is shown in Fig.1. Specifically, the magnetic field and current density distributions of the same material under different magnetic field frequencies are first analyzed. Then, a copper sphere is introduced as a comparison model to analyze the magnetic field and current density distributions of different materials under the same magnetic field frequencies, thereby further exploring the influence of parameters such as electrical conductivity and magnetic permeability on the skin effect.

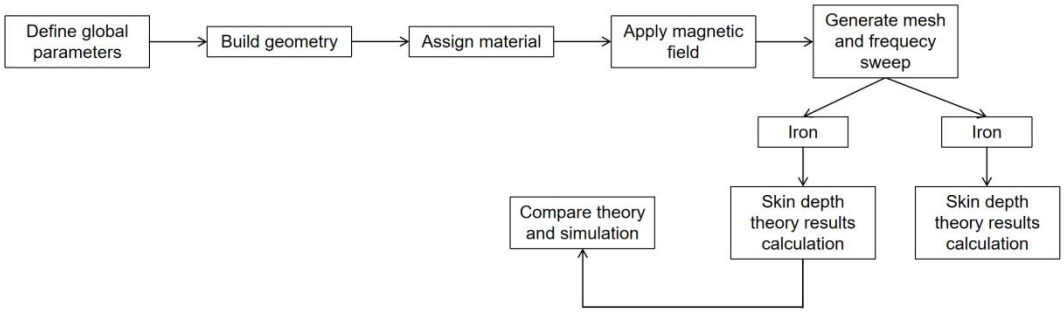


Fig.1 Experimental flow

The procedures of this lab

1. Define parameters

First, as shown in Fig.2, global physical parameters are defined in COMSOL, including the radius of the iron sphere r0 and the amplitude of the background magnetic flux density B0.

参数

名称	表达式	值	描述
r0	0.125[mm]	1.25E-4 m	Conductor sphere radius
B0	1[mT]	0.001 T	Magnetic flux density amplitude

Fig.2 Definition of global parameters

Defining global parameters helps standardize and simplify the subsequent modeling and simulation process.

2. Build geometry

Next, according to the experimental requirements (as shown in Fig.3), a three dimensional sphere model is created, where the inner region is the conductor sphere and the outer region is the surrounding air domain.

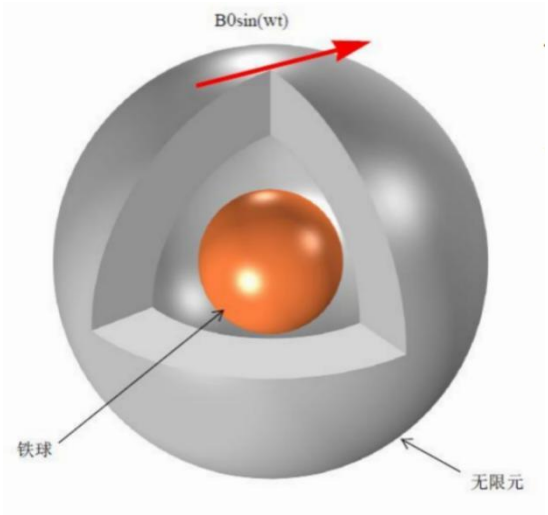


Fig.3 Geometry schematic

The model adopts a concentric-sphere configuration centered at the origin, with the z-axis as the rotation axis and the xy-plane as the working plane. The inner domain is the conductor sphere with radius r0, while the outer air domain is set to four times the sphere radius to ensure accurate representation of electromagnetic wave propagation and attenuation. The detailed parameters are shown in Fig.4.

Fig.4 Geometry parameter settings

Fig.5 shows the constructed geometry, where the inner sphere is the conductor region and the surrounding layers represent air, approximating an unbounded electromagnetic environment.

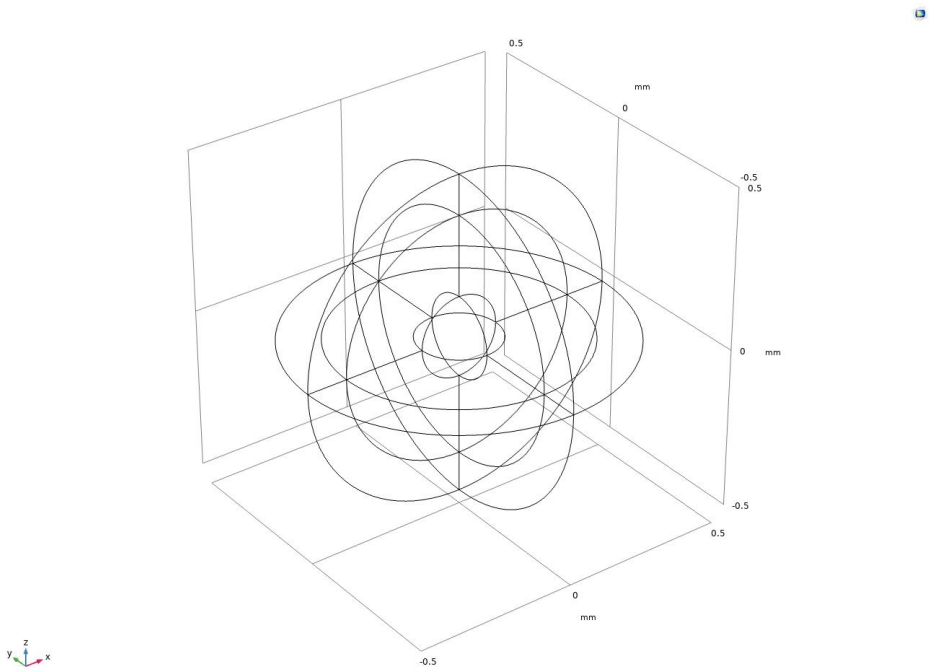
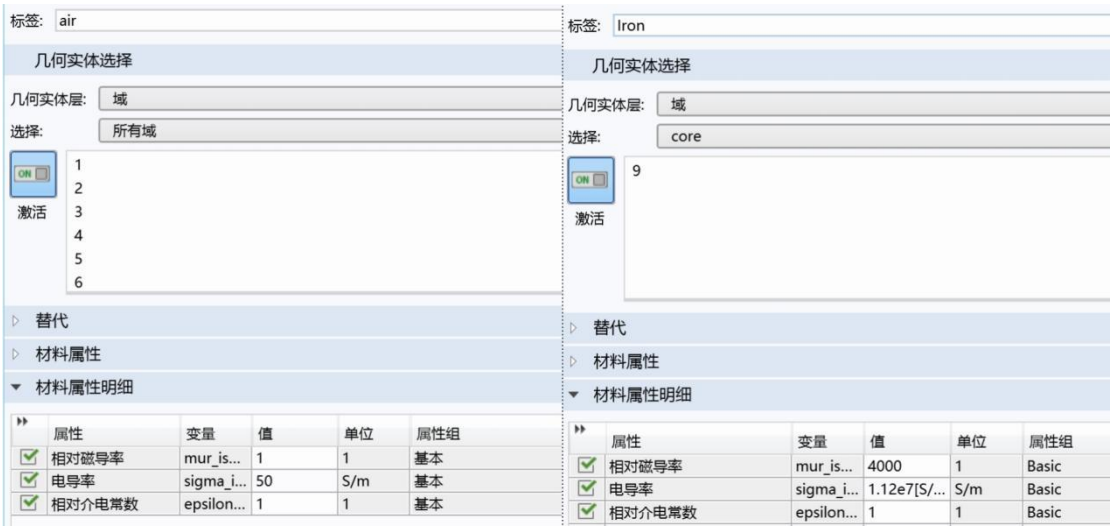


Fig.5 Geometry model in COMSOL

3. Assign materials

After building the geometry, the central sphere is assigned the material properties of iron, while the surrounding region is assigned air properties to avoid electrical conduction in the air, as illustrated in Fig.6.



(a) **(b)**

Fig.6 Material settings for the iron-sphere model

To further investigate the effect of material properties on the skin effect, a copper-sphere model is also constructed for comparison with the iron model. The material parameters are shown in Fig.7.

Fig.7 Material settings for the copper-sphere model

4. Apply background magnetic field

In this experiment, the background magnetic field is applied using the vector potential method. According to electromagnetic theory, the magnetic flux density **B** and the vector potential **A** are related by :

$$B = \nabla \times A \tag{1}$$

The external magnetic field is defined as a uniform alternating magnetic field with amplitude B0， $B = B_0 \sin(\omega t)\hat{z}$。

To satisfy Equation(1), the vector potential is chosen as $A = (0, A_z(y), 0)$. From the theoretical derivation :

$$B_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \tag{2}$$

Since $A_x = 0, A_y = 0$ in this experiment, we obtain $B_z = \frac{\partial A_z}{\partial y}$. Thus, the background vector potential can be expressed as:

$$A = (0,0,B_0 y) \tag{3}$$

Based on this derivation, a background magnetic field node is added in the magnetic field physics interface in COMSOL, and a uniform AC magnetic field in the vertical direction is applied. The corresponding settings are shown in Fig.8.

Fig.8 Background magnetic field settings

5. Mesh generation

To accurately capture the current density variation near the conductor surface under the skin effect, a free tetrahedral mesh is applied to the entire computational domain.

The core region (the iron sphere) is selected in COMSOL and assigned a local mesh control node, where an extra-fine mesh is used to ensure high resolution. The mesh settings and an example result are shown in Fig.9.

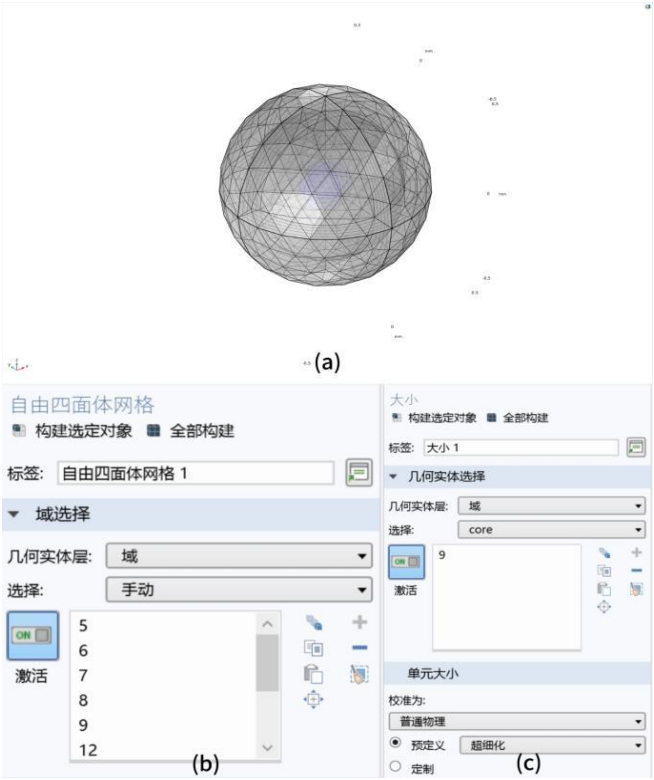


Fig.9 Mesh settings and simulation result

6. Frequency sweep

To investigate the distribution of magnetic field and current density at different frequencies, a frequency-domain solver is selected (Fig.10), and simulations are performed at f = 200 Hz, 20 kHz, 20 MHz.

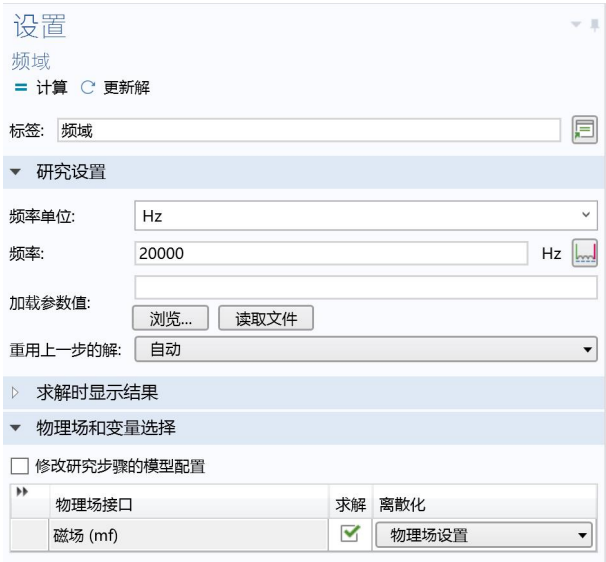


Fig.10 Frequency-domain solver settings

7. Material comparison

To further analyze the influence of electrical conductivity and magnetic permeability on the skin effect, the conductor material is changed from iron to copper, and the same simulation procedure is repeated. The material properties of iron and copper are summarized in Table 1.

Table 1 Material properties

Property	Iron	Copper
Relative permeability μr	4000	1
Conductivity σ(S/m)	1.12×10^{7}	5.98×10^{7}

According to electromagnetic theory, the high magnetic permeability of iron results in a stronger skin effect, whereas the higher conductivity of copper leads to a deeper skin depth.

8. Theoretical Calculation of Skin Depth

When an alternating current flows through a conductor, the current density tends to concentrate near the surface and decays exponentially toward the interior, a phenomenon known as the skin effect. The depth at which the current density decays to about 36.8% of its surface value is referred to as the skin depth:

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} \tag{4}$$

where f is the frequency, μ is the magnetic permeability, and σ is the electrical conductivity.

From Equation(4), it follows that the skin depth decreases as the frequency increases. By substituting the frequencies and the material parameters listed in Table 1 into Equation(4), the skin depth for each material at different frequencies can be obtained, enabling an analysis of the role of material parameters in the skin effect.

Results and discussions

1. Simulation and Calculation Results of the Iron Sphere

Based on Equation(4) and the relevant parameters, the skin depth of the iron sphere at different frequencies was calculated. The **results** are shown in Table 2.

Table 2 Skin depth of the iron sphere (theoretical results)

Frequency	Skin depth δ(Iron)
200 Hz	168.14 μm
20 kHz	16.81 μm
20 MHz	0.53 μm

The simulated current density distributions of the iron sphere under 200Hz、20kHz 和 20MHz magnetic fields are shown in Fig. 11, Fig. 12, and Fig. 13, respectively.

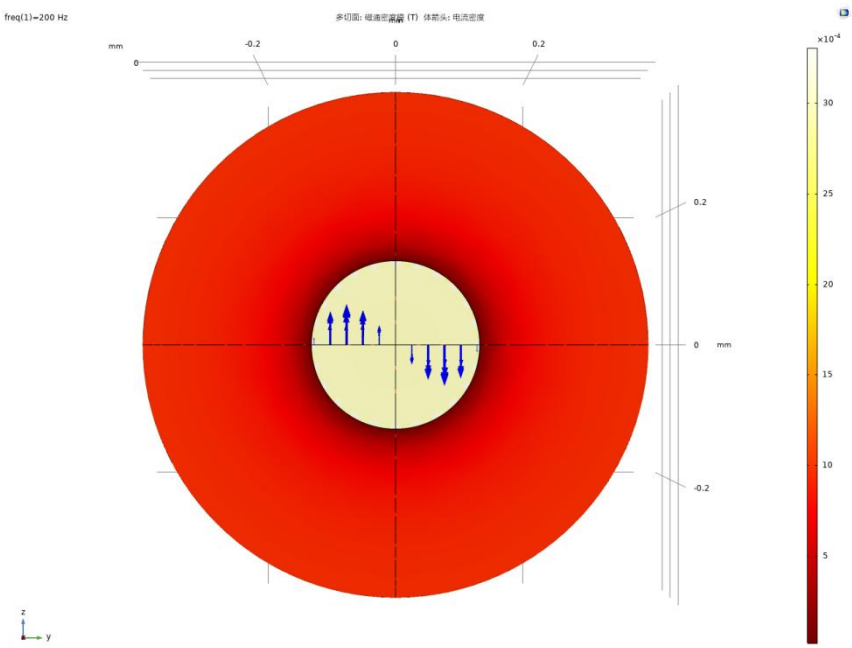


Fig. 11 Simulation result of the iron sphere at 200 Hz

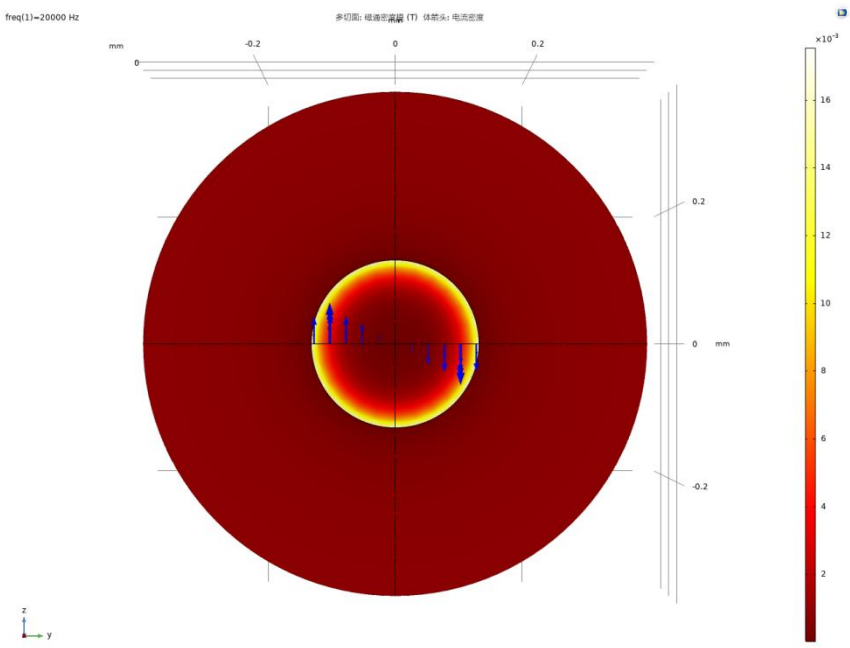


Fig. 12 Simulation result of the iron sphere at 20 kHz

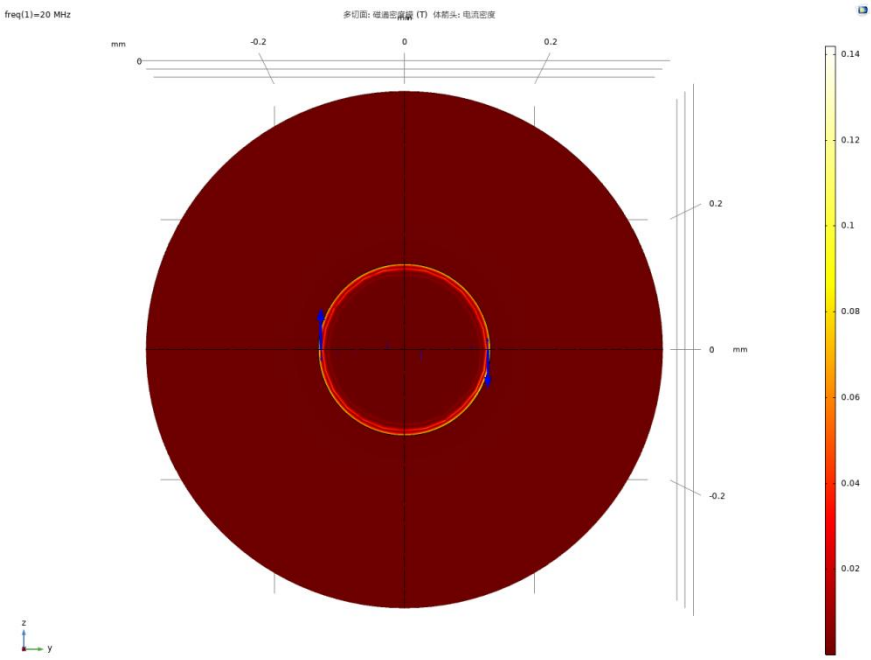


Fig. 13 Simulation result of the iron sphere at 20 MHz

It can be observed that at 200 Hz, current penetrates into the interior of the iron sphere, and the skin effect is weak. At 20 kHz, current becomes concentrated around the surface, and the current density inside the conductor decreases significantly. At 20 MHz, current is almost completely confined to the surface layer, and the current inside the conductor approaches zero. These results indicate that, for the same material, the skin effect

becomes more pronounced with increasing frequency, and the skin depth decreases accordingly.

2. Simulation and Calculation Results of the Copper Sphere

According to Equation(4) and the relevant parameters, the skin depth of the copper sphere at different frequencies was calculated, **as shown in Table 3.**

Table 3 Skin depth of the copper sphere (theoretical results)

Frequency	Skin depth δ(Copper)
200 Hz	4.602 mm
20 kHz	0.4602 mm
20 MHz	0.01455 mm

The simulated current density distributions of the copper sphere under 200Hz、20kHz 和 20MHzmagnetic fields are shown in Fig. 14, Fig. 15, and Fig. 16, respectively.

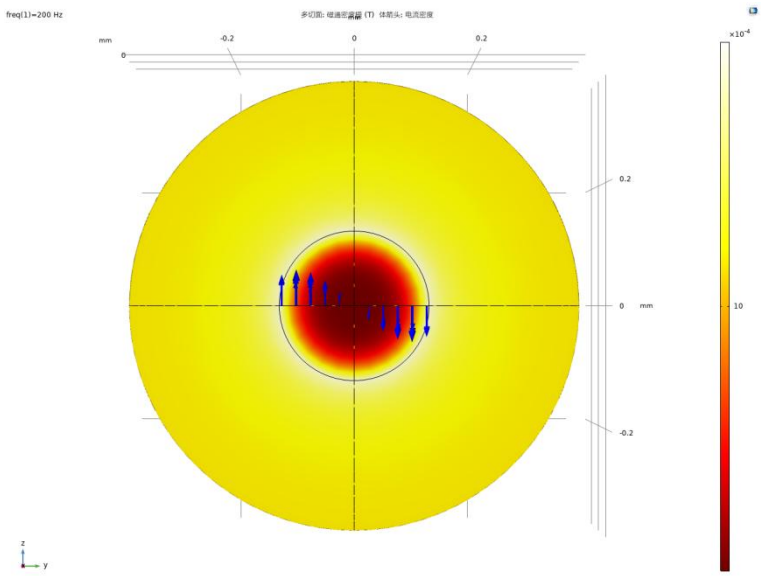


Fig. 14 Simulation result of the copper sphere at 200 Hz

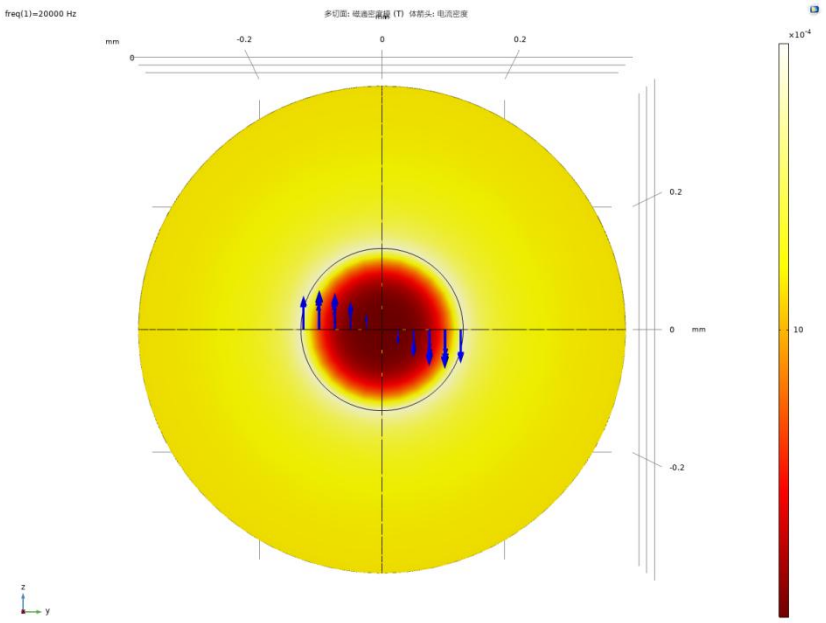


Fig. 15 Simulation result of the copper sphere at 20 kHz

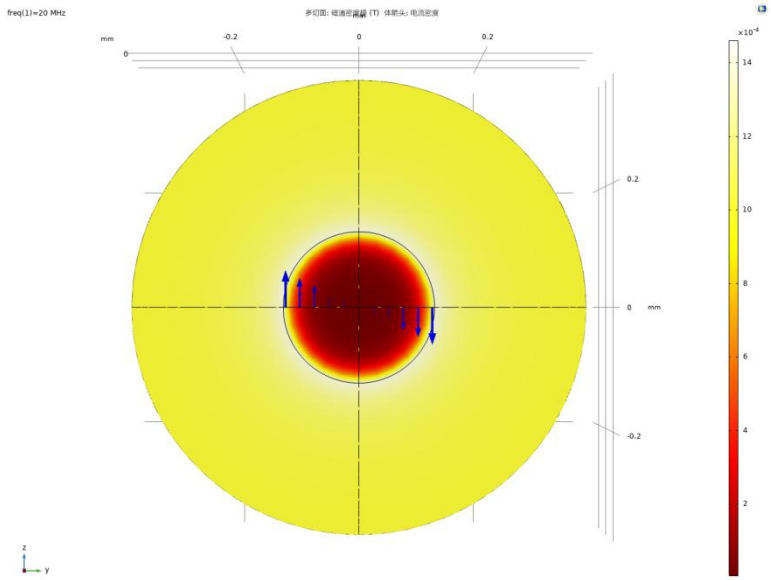


Fig. 16 Simulation result of the copper sphere at 20 MHz

It can be seen that at 200 Hz, current almost fully penetrates the copper sphere with no significant attenuation inside. At 20 kHz, current still penetrates the conductor with only slight attenuation near the surface. At 20 MHz, however, current becomes concentrated near the surface, and the skin effect becomes pronounced. Similar to the iron sphere, the skin effect in the copper sphere increases with frequency, and the skin depth correspondingly decreases.

3. Comparative analysis

A comparison of the simulation results of iron and copper at the same frequencies shows that the skin depth of the iron sphere is significantly smaller than that of the copper sphere, due to the much higher magnetic permeability of iron.

Specifically, at low frequency (200 Hz), the skin depth in the iron sphere is comparable to the sphere radius, while the skin depth in the copper sphere is much larger than the sphere radius, allowing current to distribute almost uniformly inside the copper sphere. At high frequency, current in the iron sphere is confined to a very thin surface layer, whereas in the copper sphere, although surface current concentration appears, some current can still penetrate into the conductor interior.

In summary, due to its high magnetic permeability, the iron sphere exhibits a stronger skin effect compared to the copper sphere. The copper sphere, with higher electrical conductivity, only shows significant skin effect at high frequencies. This agrees well with the theoretical skin-depth formula.

Conclusions

This experiment investigates the skin effect of conductors under alternating magnetic fields through theoretical calculations and COMSOL simulations, and analyzes the influence of frequency and material parameters on skin depth and current distribution. The main conclusions are as follows:

1. **Higher frequency leads to a stronger skin effect**: At low frequencies, current distributes throughout the conductor interior and the skin effect is negligible. As the frequency increases, current inside the conductor decays more rapidly and gradually concentrates near the surface, becoming almost entirely confined to the surface layer at high frequencies.

2. **Greater magnetic permeability results in a stronger skin effect**: Materials with higher magnetic permeability exhibit stronger magnetic field response inside the conductor, causing faster attenuation of current and more pronounced surface current

concentration, thus leading to a stronger skin effect.

3. **Higher electrical conductivity results in a weaker skin effect**: Materials with higher electrical conductivity show little skin effect at low and moderate frequencies; however, noticeable surface current concentration still occurs under high-frequency conditions.

In summary, this experiment demonstrates the influence of frequency, magnetic permeability, and electrical conductivity on current distribution in conductors, providing both theoretical and simulation-based support for engineering applications such as high-frequency conductor design and analysis of conductor behavior in high-frequency electromagnetic fields.